

PLAYER AID

BY GEOFF SPEARE

HIGH FRONTIER
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Player Turn: Move Spacecraft, any # of Free Actions, 1 Operation, in any order.

Anchored Home Bernal: Gain 1 Aqua at start of Turn.

Operations

● Glitch Trigger ☹ Felony Blue text indicates rules from Modules 0, 1, and/or 2.

Operation	Summary	Details
Income	Get 1 Aqua	• Politics: This Operation replaced by the Fundraise Operation
Research Auction	Auction top card from any deck.	• Must have < 4 cards in Hand to start or to bid. • Auctioneer wins ties; pay auctioneer (Pool if auctioneer wins). • For each required support, take top card from that deck. • Bernal: With a Home Bernal, 2nd Bernal starts in Bernal Stack.
Free Market	Sell card from Hand or Black-Side card from LEO Stack.	• Get 3 Aqua for Hand card. • Get card's Exploitation Track value in Aqua for Black-Side card.
Boost	Play White-Side cards from Hand to LEO (or Home Bernal).	• Pay Mass in Aqua. • Set radiator Orientation. • To Home Bernal: x2 cost.
Site Refuel	● Add Fuel to Rocket, or FT to Stack.	• ISRU Refuel: Hydration – ISRU + 1 = water (<i>see separate table</i>). • Factory Refuel: 7 water / 1 isotope (GW) of Factory type; – Felony ☹ at opponent's Factory. – Bernal: Dirtside fuel can go directly to Anchored Bernal. • Air-eater: Pac-Man ☹ on Aerobrake: water FTs = 5 - fuel consumption.
Prospect	● Evaluate and claim Site(s) (<i>see separate table</i>).	• ISRU ≤ Site Hydration (<i>see Prospecting Table</i>). • Roll 1d6: if ≤ Site Size, successful; or Site is Busted. • Raygun: prospect any/all Adjacent non-Atmospheric Sites. • Buggy: re-roll once OR prospect all Sites connected by buggy roads.
Industrialize	● Create a Factory.	• Decommission <i>any</i> robonaut and refinery (plus supports) at claimed Site. • Radiator supports not required and not Decommissioned.
ET Production	Produce Black-Side card from Hand to a Factory.	• Must match Spectral Type of Factory Site (C/S/M/V/D/H). • Set radiator Orientation. • Bernal: Dirtside ET production can go directly to Bernal. • Colonists: If building a Robot, must Discard a Colonist.
Delivery	● Move Black-Side card from Factory to LEO / Home Bernal	• Pay FT from Factory = zones from Earth x2 (+1 if Site > 6 Size). • Freighters: This Operation replaced by Freighter cards.
Fundraise (Politics)	Place/move delegate; gain Aqua.	• Gain 1 Aqua, play delegate; move delegate to Adjacent Ideology. • May place on your color Ideology or where you have a delegate. • Perform vote tally (count votes, change Active Law). • Politics: If martial law is Active, Discard 1 opponent delegate.
Promotion (Freighters, GW Thrusters, Colonists)	Flip card to Purple-Side.	• Card must be at (Bernal: Anchored above) its Promotion Site. • Felony ☹ at Opponent's Factory. • Bernal: Card can be at Promoted & Anchored Bernal instead. • Bernal: Exomigrate if a Bernal was Promoted.
Nanofactory (Freighters + Bernal)	Create Mobile Factory at Anchored non-Home Bernal.	• Must have Promoted Freighter. • Decommission robonaut, refinery, supports including radiators.
Anchor (Bernal)	● Create space station; gain Bernal Ability.	• Bernal must be in Home Orbit or non-Site, non-Hazard Space adjacent to Factory. • Decommission Bernal supports, gain Bernal Ability. (Solar-Powered Cards don't work past Ceres Zone without Powersat.) • Place Colony Dome on Bernal (now considered Human). • Perform exomigration free action immediately.
Homesteading (Bernal)	Build Colony (at Factory without a Colony).	• Discard Black-Side card at LEO or Home Bernal. • Must also Decommission a Human Colonist (anywhere) and exomigrate.
Epic Hazard (Freighters / Futures)	Complete Future / Build Space Elevator.	• Future: Requires Colocated Crew or Colonist and Future Card. • Elevator: Requires Colocated Freighter or Mobile Factory. (Earth GEO Space Elevator is only built via a Bernal Ability.) • Epic Hazard Roll, fail on a ☐ – Decommission unit used for roll. • Elevator: Claim Unclaimed/Busted/Jumpable Sites at both ends.

Free Actions

● Glitch Trigger ⚖️ Felony Blue text indicates rules from Modules 0, 1, and/or 2.

Free Action	Summary	Details
Cargo Transfer (Glitch Trigger when moving to 2 nd Stack) ●	Move cards and/or FTs between Colocated Stacks.	- Can form a new Rocket / Outpost Stack. - May Decommission (⚖️ for Humans). - Refuel ∞ dirt FTs (1 FT for crew thrusters) with any ISRU card at Factory or Site. - Freighters: Check Load Limit & Factory Load. - Bernal: ∞ dirt FTs at Anchored Bernal.
Internal Tankage	Convert between FTs and Fuel.	Can Decommission cards for dirt fuel.
Build Colony	Create permanent Colony at Factory.	- Decommission Crew/Colonist at Factory. - Colonists: Exomigrate if a Colonist was used. - Politics: Place 1 delegate in centrist or Ideology of card used; perform vote tally.
Claim Jump ⚖️	Replace opponent's Claim with yours.	- You must have a Human present. - No opposing Factory or Human present. - Futures: Always allowed on Sites specified on Futures you own.
Load Glory Chit	Load / Claim glory chit from Site previously unvisited by Humans.	- Humans must be present. - Each Crew or Colonist can carry 1 glory chit at a time.
Voluntary Discard (Felony for Humans) ⚖️	Discard cards and/or figures (1 Human per Turn max).	- Discard cards to bottom of patent decks. - Remove figures if their Stack is empty.
Glitch Repair	Remove Glitch token from Stack.	Human must be Colocated with Stack.
The Martian	Move Crew or Colonist along buggy road.	Requires Operational buggy.
Lobby (Politics)	Remove delegate to gain inactive Law benefits.	- Once per Turn. - Pay 1 Aqua (free in War); remove 1 delegate.
Big Cube Swap (Freighters)	Swap Freighter cube with Factory cube.	Only if Freighter is Promoted and not carrying any cards.
Exomigration (Colonists) Mandatory!	If below Colonist limit: Gain topmost Colonist to LEO or Home Bernal (or Robot to Hand).	- Limit: 1 per Anchored Bernal (+1 if Promoted). - If queue is empty, emancipate Robots. - Robots go to Hand; exomigrate again. - Politics: place 1 delegate (based on Colonist's loyalty); perform vote tally.
Unanchor (Bernals)	Anchored Bernal becomes unanchored and can move again.	- Promoted status does not change. - Remove Colony dome & 1 Colonist (2 Colonists if unanchoring Promoted Bernal). - Can perform a free dirt refuel.
Space Elevator (Freighters)	Move between ends of a Space Elevator.	- Connected Factory owner's permission required. - Elevator must be built via Epic Hazard Operation.

Prospecting

- 1) Check Site Hydration and robonaut ISRU on below table to see if prospecting is possible.
- 2) Roll less than or equal to Site Size on 1d6 to establish a Claim (Size > 5 = automatic success).

Prospect Operation	💧💧💧	💧💧	💧	💧	No Hydration
ISRU 4	Can prospect (1)	NO	NO	NO	NO
ISRU 3	Can prospect (2)	Can prospect (1)	NO	NO	NO
ISRU 2	Can prospect (3)	Can prospect (2)	Can prospect (1)	NO	NO
ISRU 1	Can prospect (4)	Can prospect (3)	Can prospect (2)	Can prospect (1)	NO
ISRU 0	Can prospect (5)	Can prospect (4)	Can prospect (3)	Can prospect (2)	Can prospect (1)

ISRU Refuel Operation: Refuel # of Water FT equal to number in parentheses above based on ISRU/Hydration.

Felonies

Felony	Event	Defense
Claim Jump (Free Action)	Replace opponent's Claim disk with yours or claim Luna. (Futures) Always allowed for Sites specified on Futures you own.	Opposing Humans present
Factory Hijack	Refuel, factory-assist landing/liftoff, or Promotion using opponent's Factory.	Opposing Humans present
Murder / Suicide	Decommission your Humans (once / Turn) or colonize the kreutz sungrazer. (Politics/Colonists) If this Decommissions a Colonist, remove a delegate.	N/A

You must have a Human present to commit any Felony.

Thrust

Dry Mass: total Mass of all Rocket components/Cargo

Wet Mass: Dry Mass plus fuel (cannot be less than Dry Mass)

Fueling or adjusting Mass: use red arrows.

Consuming fuel: move fuel marker on black arrows.

Isotope Fuel (GW): Fuel matches Spectral Type of GW Thruster. If thruster is lost, Isotope Fuel reverts to water FTs.

Promotion: indicates Colony type required for promotion op.

Pac-Man: Can perform air-eater refuel operation

Ballerina: Each icon provides 1 Bonus Pivot.

Base Thrust: Max # of Burns/ Turn, Landing & liftoff limit (modified)

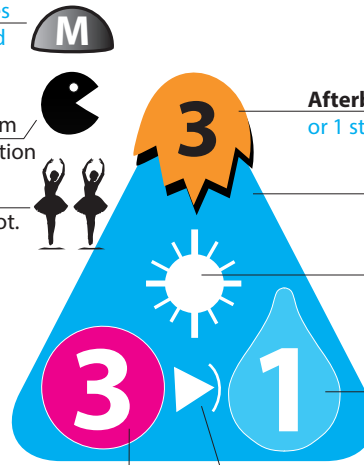
Pushable: Can use beamed power modifier.

Afterburn: spend X steps of fuel for +1 thrust or 1 step of isotope fuel for +X thrust (once / Turn).

Fuel Grade: water (blue), dirt (gray), isotope (yellow)

Solar Power: Heliocentric modifiers apply to this Spacecraft.

Fuel Consumption: # of fuel steps used per Burn



Spacecraft Movement

● Glitch Trigger ☞ Felony Blue text indicates rules from Modules 0, 1, and/or 2.

Movement steps	Details
1. Activate an Operational thruster	Each Rocket can only use one thruster per Turn.
2. Starting net thrust = base thrust	Value in lower left corner of triangle.
3. Modify for afterburn	If thruster has afterburn at top, may spend cost for +1 net thrust (once/turn). GW: GW/TW thrusters spend 1 step of fuel for listed thrust bonus. (Afterburning provides 1 Therm of cooling for the thruster support chain.)
4. Modify for Wet Mass	+2 to -2 based on weight class (Wet Mass after adjusting for afterburn).
5. Modify for Heliocentric Zone	Only if thruster or its supports (see #7) have solar icon ☼.
6. Modify for beamed power	☞ +1 if thruster has push icon and player has beamed power / Powersat.
7. Modify for supports	☞ Net thrust, fuel consumption, and/or solar icon ☼ (round fuel consumption fractions up). Ignore movement-modifiers for supports needed for radiators, reactors, Freighters, or GW/TW thrusters. Solar-Power supports ☼ can never be used in the Neptune Heliocentric Zone.
8. Result is net thrust	Net thrust will not change even if Rocket composition changes mid-move (e.g. Cargo is destroyed).
9. Note Bonus Pivots	☞ Pivots which can be performed without requiring Burns.
10. Note fuel consumption	Fuel consumption value in lower right corner of triangle.
11. Move	Move through Pivots and Burns until # of Burns (or fuel) are used up; then Coast as far as desired (including Landing).

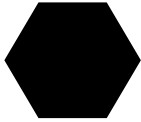
Each Burn expends a # of fuel steps equal to the fuel consumption; Net thrust = max # of Burns per Turn.

Freighter Movement

Movement steps	Details
1. Use Freighter Movement Rules	Freighter Stacks cannot use other thrusters.
2. Starting Net Thrust = 1	All Freighters have a base thrust of 1.
3. +1 for any type of beamed power	☞ Only if player has beamed power / Powersat.
4. Result is Net Thrust	This will be 1 (+1 if using beamed power), may be modified by Abilities.
5. Note Bonus Pivots	☞ Pivots which can be performed without requiring Burns.
6. Move	Move through Pivots and Burns until # of Burns is reached; then coast as far as desired (including landing).
Freighters do not track Fuel	–
Promoted Factories move as Freighters	If Mobile Factories enter/leave a Claim, add/remove a Factory. They can't leave a Colony.
Freighter Landing / Liftoff	1. Land or liftoff of Size 1 Sites. 2. Land or liftoff of Size 1-5 Sites with factory-assist / rocketplane (H6c). (Mobile Factories can use themselves for factory-assist).
Freighter Radiation and Flares	☼ If card in Stack would be Decommissioned from Radiation, Stack is Glitched instead (no effect if Stack is already glitched).

Solar System Map

● Glitch Trigger ☹️ Felony Blue text indicates rules from Modules 0, 1, and/or 2.

Name	Icon	Restrictions
Hohmann Intersections (no circle)		- Freely move straight through (if possible) or stop. - Must Pivot (2 Burns) to change direction. - If begin Turn on Hohmann Intersection, free to move in any direction.
Lagrange Point Intersection (circle)		Freely move straight through, stop, or change direction.
Lagrange Burn		1 Burn to enter.
Lander/ Half Lander Burn		1 Burn to enter Lander. 1 Burn (½ fuel consumption) to enter Half Lander. - Cannot end move on these Spaces.
Site (Land and Liftoff) (☹️ using opponent's Factory)		Free to land or liftoff if net thrust > Site Size with the following exceptions: Aerobrake path landing: Hazard Roll required. Factory-assist landing or liftoff: Hazard Roll required (unless you have Powersat) and cannot move to/from lander burns. Rocketplane liftoff (H6c), can use factory-assist as above for Atmospheric liftoff (pay 2 * Wet Mass in FTs).
Synodic Comet Site (Land and Liftoff)		As above for Sites; additionally, cannot enter or exit unless Sunspot Cube is in matching-colored season. (GW: TW Thrusters can enter/exit in any season.)
Flyby	● 	- Gain Bonus Burns which don't use fuel (can't be used for lander burns). - Must have activated, operational thruster.

Threats

Threat	Triggers	FINAO	Roll Effect
Hazard	Aerobrake Hazard Spaces; Crash Hazard Spaces; Factory-Assist Landing/Liftoff 	Yes	1d6: Decommission Stack on a ☐.
Glitch ●	Flybys; Cargo Transfer; Site Refuel; Prospect; Industrialize	No	1d6: Decommission collocated cards with matching rad-hardness; roll before triggering event occurs.
Belt	Radiation Belt Spaces 	No	1d6, subtract net thrust, +2 if in Sunspot Cycle in red: Decommission cards with lower rad-hardness (Freighter Cargo: Stack receives a Glitch token instead).
CME	Solar Flare Event (roll of ☐ or ☐ during red Sunspot Cycle)	No	1d6 + Heliocentric Zone modifier: Decommission cards with lower rad-hardness (LEO, Cycler, non-Sol Belt Spaces, Sites, Home Bernals unaffected).
Epic Hazard (Futures)	Epic Hazard Operation	Yes	1d6: Decommission all Crew/Colonists present and Future is not attained on a ☐.

FINAO = Failure Is Not An Option; spend 4 Aqua before roll to ignore Hazard.

Events

Roll	Event	Effect
☐ or ☐	Inspiration	Put top card of each deck & Colonist queue on bottom of that deck.
☐	Glitch	Each player places Glitch token on glitch-free non-LEO Stack with most cards.
☐	Pad Explosion / Space Debris	Each player Decommissions heaviest card in LEO. (Crew, Black-Side, Purple-Side, Bernal, Colonist cards and Mobile Factories are immune.)
☐ or ☐	Anarchy (Blue)	- All Faction privileges lost, all players have Felonious for rest of blue season. - Politics: Roll 1d6, remove 1 delegate of each player from area rolled; no Law is Active (marker still moves, all Laws may be lobbied).
	Budget Cuts (Yellow)	Each player Discards a card from Hand.
	Solar Flare (Red)	Make a global CME Roll (see CME Threat above).

Game Setup

● Glitch Trigger ⚔️ Felony Blue text indicates rules from Modules 0, 1, and/or 2.

Module	Notes	Starting Aqua	Card Limits
Core	Base game	6	
Politics (Module 0)	Assembly placard & starting delegates in play; Factory cubes used for player delegates		
Freighters (Module 1)	Freighter deck in play	+1	1 Freighter
GW Thrusters (Module 1)	GW Thruster deck in play	+1	1 GW Thruster
Futures (Module 1)	End game goals for VP and effects		
Bernal (Module 2)	Bernal deck in play; Faction privileges inactive until Home Bernal acquired	+1	2 Bernals
Colonists (Module 2)	Colonist queue in play		

Game Duration / Scoring

Game Duration: 4 Seniority Disks, if GW, Freighters, or Colonization: 4 or 5 Disks, if Futures: 7 Disks.

Politics: 1st Player performs vote tally after final turn including Legacy Votes, see *final vote* (O6b).

Scoring	Points
Token VP	1 VP for each wooden/plastic Token (Rocket, Claim disk, Factory cube, Colony dome, Bernal) on map or assembly (Politics).
Bonus VP	Factory: +8 / +5 / +4 VP each, according to Exploitation Track or 2 VP if in War in non-solo game (Futures) (Freighters: x2 VP if Factory is at a Space Elevator).
	Colonies: +1 VP on Astrobiology; +2 VP on Submarine or Bernal.
	Glory chits: VP per chit based on which side is faceup.
	Dirtside Hydration (Bernals): 1 VP for each Dirtside Hydration (include other players' Factories). (Bernals in Home Orbit have +6 Dirtside Hydration.)
	Futures: VP as listed on each completed Future.
Politics VP (use Active Law after Final Vote)	Freedom: +1 VP per Factory cube.
	Honor: +1 VP per glory chit.
	Unity: 1 VP per Ideology you have a delegate in.
	Authority: +1 VP per Claim disk.
	Equality: +1 VP per Colony dome.
	Individuality: +1 VP per wood/plastic token on Sites with Hazardous lander burns.
Bernal VP (for Anchored/Promoted Bernals)	Cancer Hospital Bernal (Promoted): +1 VP per Colony dome.
	Climate Control Bernal (Promoted): +2 VP per Dirtside for this Bernal.
	SSO Diplomatic Bernal: +1 VP per delegate in your color Ideology.
	SSO Diplomatic Bernal (Promoted): +1 VP per delegate.
	Tourism Cyclor (Promoted): +2 VP per Dirtside for this Bernal.

Module 0 – Politics

Summary	
- Delegates vote for an Ideology, each of which has a Law which creates a different game effect. - Place and/or move delegates with the fundraise operation, which can change the Active Law. - Use the lobby free action to remove a delegate and get an inactive Law's benefit once. - Receive VP for delegates and for bonuses based on the Active Law at game end.	
Rules Changes	Law Summary
Fundraise Operation replaces income operation. New Free Action: Lobby. Cubes serve as delegates and/or Factories. Vote Tally: Check for new Active Law (based on most delegates) after delegates placed. Legacy Votes: Place removed Seniority Disks on an Ideology (1st Player chooses which). 1st player chooses new 1st player. Final Vote: Count delegates and legacy votes at end of game to determine final Active Law for scoring.	Freedom: Sell 2 cards for 5 Aqua with free market op. Honor: Gain 1 Aqua per glory chit with fundraise op. Unity: Every Law with 2+ delegates is Active; no lobbying. Authority: After vote tally of fundraise op, may remove 1 delegate. Equality: Research auction operation skips the auction; take 1 card for 1 Aqua (without listed supports or hand limit restrictions). Individuality: Treat Factories and Bernals as your own for all non-victory purposes. Centrist: During pad explosion / space debris event, gain Aqua equal to Mass of Decommissioned card. (Lobby during event resolution).

Module 1 – GW Thrusters, Freighters, Futures

Summary	
GW Thrusters: Powerful thrusters which require special Isotope Fuel. Freighters: Slow-moving Cargo carriers; can make all your Factories movable. Futures: Private quests gained from promoting cards to Purple-Side to gain VP and abilities. Requires Modules 0, 1, and 2	
Rules Changes	Mobile Factory Summary
• Delivery Operation no longer allowed. • New Operations / Free Actions: Promotion, Nanofactory, Epic Hazard, Big Cube Swap. • Futures. Optional during setup and require Modules 0, 1, and 2. On-Board Supports: Some Freighters provide supports. Special Futures: • Ad Astra Futures have special requirements. • Casus Belli Futures lead to War of Independence. Major game changes (<i>see glossary!</i>) • Robot Emancipation: Treat Robots as Humans.	• Once Freighter is Promoted, all your Factories become Mobile. • Same characteristics as Freighter except Cargo and Futures. • Cannot leave a Colony without a Factory/Freighter. • Are not Factories when not on a Claim. • Can be created at a Bernal via the nanofactory operation. • Swap Freighter and any Mobile Factory via big cube swap free action. • If Freighter is Decommissioned, any Mobile Factory can become the Promoted Freighter's new location. Promotion: Flip cards to Purple-side at specific Colony Sites (<i>see Module 2 Summary</i>).

Module 2 – Bernals, Colonists

Summary		
Bernals: Large mobile space stations which can be Anchored in Home Orbits near Earth or Adjacent to Factories (Dirtsides). Colonists: Space settlers who lend their abilities to your efforts as your infrastructure grows. Futures: Private quests gained from promoting cards to Purple-Side to gain VP and abilities. Requires Modules 0, 1, and 2.		
Rules Changes	Bernal Rules	
New Operations / Free Actions: • Promotion, Homesteading, Anchoring, Exomigration • Factory Refuel and ET Production from Dirtside can go straight to the Bernal. • Build Colony Free Action: May Decommission a Colonist instead of Crew. Faction Privileges Locked: Inactive at start of game. On-Board Supports: Some Colonists provide supports. These Colonists can be Decommissioned as needed supports for industrialization - this is not a Felony. Colonist Queue: Colonists are not a patent deck and enter play via a new mechanism (exomigration). Decommissioned Colonists go back to the queue. You must Exomigrate if below Colonist Limit. Promotion: Purple-side cards must be Promoted at specific Colony Sites or a Promoted, Anchored Bernal as listed below.	When A Home Bernal is Anchored: • Faction privilege unlocked. • Bernal "Home" Ability (if any) unlocked. • Gain 1 Aqua at start of each Turn. • Store Aqua and Crew at Bernal instead of LEO . (move them from LEO when Home Bernal created) • 2nd Bernal purchased starts in Bernal Stack instead of Hand. • All LEO actions may be performed at Home Bernal instead (free market, delivery, etc.). • May boost directly to Bernal for 2x cost (skip Belt Roll). • Colocated cards immune to Glitches, Pad Explosion / Space Debris, Solar Flares. Bernal Special Exceptions: • Bernals must land on Synodic Comet Sites to use them as Dirtsides. • Bernals cannot use Luna Sites as Dirtsides. • Bernal Abilities are not active until Anchored. Bernal Colonist Limit: • 1 Colonist per Bernal, +1 if Anchored. Add or lose Colonists whenever a player's limit changes.	
Promotion Type	Colony Site	Colonist Specialties
	Spectral Type: C S M V D H	Miner: 1 free Colocated Site Refuel per Turn per miner. Prospector: 1 free Colocated Prospect or Promotion per Turn per prospector. Industrialist: 1 free Colocated Industrialize, Anchoring or Nanofactory per Turn per industrialist. Engineer: Colocated ET Production can build 1 extra card per engineer present.
Submarine		
Astrobiology		
Atmospheric		
Sunbelt		
TW Thruster, Freighter, and Colonist cards can always be Promoted at an Anchored & Promoted Bernal		